SUPERSEDING
Int. Fed. Spec. P-S-00606c (Army-GL)
December 27, 1962 and
Fed. Spec. P-S-606a
January 28, 1942

FEDERAL SPECIFICATION

SOAP, SCRUBBING, ALKALINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers alkaline scrubbing soap in powder form (see 6.1).

1.2 Classification. The scrubbing soap shall be of one type and one grade.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-D-723—Drums, Fiber.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 536—Soap and Soap-Products (Including Synthetic Detergents); Sampling and Testing.
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The alkaline scrubbing soap shall consist of soap and alkaline salts which

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have been mixed in the molten state before crystallization. The soap shall dissolve readily in tepid water.

3.2 Composition. The alkaline scrubbing soap shall conform to the requirements specified in table I when tested as specified in table II.

TABLE I.—*Composition*

Characteristic	Percent by weight	
	Minimum	Maximum
Anhydrous soap	15.0	25.0
Alkaline salts (calculated as Na_2CO_3)	40.0	50.0
Sum of anhydrous soap and alkaline salts (alkaline salts calculated at Na_2CO_3)	55.0	75.0
Matter insoluble in water ..	—	1.0
Free alkali as NaOH	—	0.5
Inert fillers[1]	None	None

[1] A certificate of compliance with this requirement shall be furnished the contracting officer by the contractor. The certificate of compliance shall be accompanied by actual test, inspection or other verifiable quality data.

3.3 Color. The soap shall have a uniform color. Unless otherwise specified (see 6.2), the color of freshly sampled soap shall not be darker than color number 30257 of Fed. Std. 595 (see 6.3).

3.4 Odor. The soap shall be free of an objectionable odor. An objectionable odor, in this specification, is defined as a rancid, fishy, or horsey odor.

3.5 Form. The soap shall be in powdered form.

3.6 Labeling. All packages and unit containers shall bear the following precautionary label:

CAUTION: MAY CAUSE BURNS

Do not get in eyes. Avoid prolonged or repeated skin contact.

Do not take internally. In case of contact with eyes, immediately flush with water for at least 15 minutes and get medical attention.

3.7 Workmanship. The finished product shall be clean, uniform, free flowing, and shall not lump or cake.

3.8 Quality. The quality of the finished product shall conform to the levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQLs) specified in section 4 for any examinations and tests of materials, components, and end items, as applicable.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Sampling for inspection.* Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all soap manufactured using the same components and submitted for delivery at one time.

4.2 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that all materials and components used in the end item were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

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4.3.1 *Examination of the end item.* The end item shall be examined for the defects in the applicable sub-paragraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.6. Random samples shall be drawn from each lot of end items offered for inspection. The lot size for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of the specified capacity for examination in 4.3.2, 4.3.3 and 4.3.4 and in units of shipping containers for examination in 4.3.5.

4.3.2 *Examination of the primary container for visual defects in construction and markings.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Construction	Not in container specified.
	Any leakage or sifting of contents.
	Any split, break, dent, hole, tear, puncture, or other defect which may affect serviceability.
	Seams or closures not as specified in liner or container.
Markings (label or printing)	Omitted, illegible, incorrect, incomplete, not in accordance with requirements.
	Label or printing not clear, smeared.

4.3.3 *Examination for defects in net contents.* The sample unit for this examination shall be one filled primary container. The lot shall be unacceptable if the average net contents per container is less than the specified or indicated quantity.

4.3.4 *Examination of the alkaline scrubbing soap for defects in workmanship, color and odor.* The sample unit for this examination shall be the contents of one filled primary container.

Examine	Defect
Workmanship	Not in powdered form.
	Not clean; not uniform; any foreign matter.
	Not free flowing.
	Lumping or caking.
Color	Not uniform.
	Darker than color chip number 30257, unless otherwise specified.
Odor	Any objectionable odor (rancid, fishy, or horsey odor).

4.3.5 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, count, and markings comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Closed shipping containers fully prepared for delivery shall be examined for closure defects. This examination does not apply when the primary container is the shipping container.

Examine	Defect
Packaging (as applicable)	Not in accordance with contract requirements.
	Unit package not packaged as specified.
	Packaging material not as specified.
Packing	Not in accordance with contract requirements.
	Container not as specified; closures not accomplished by specified or required methods or materials.
	Any nonconforming component; component missing, damaged or otherwise defective affecting serviceability.
	Inadequate application of components: such as incomplete closure of case liners; container flaps, loose or inadequate sealing, strapping or stapling.
	Bulged or distorted containers.
Count	Number of packages per shipping container less than the specified or indicated quantity.
Markings (exterior and interior, as applicable)	Omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.3.6 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.2	L-8	2.5
4.3.3	L-4	—
4.3.4	L-5	1.5
4.3.5	L-4	4.0

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4.3.7 *Testing of the soap powder.* The soap shall be tested for the characteristics in table II for each lot presented for examination. The sample unit, for purposes of testing, shall be a 1-pound composite sample selected in accordance with 4.3.8. Unless otherwise specified in the contract or order, duplicate determinations shall be performed on the composite sample. The average of the 2 determinations shall comply with the specified requirements for each characteristic for lot acceptability.

4.3.8 *Composite sample.* When 5 or more containers comprise the shipping lot, a sample of not less than 1 pound shall be taken from each of not less than 5 separate shipping containers, randomly selected throughout the lot, presented for examination. When less than 5 shipping containers comprise the lot, the total samples shall, in all cases, consist of not less than 3 sample portions taken at random from the containers or container. The samples shall be placed in a receptacle of suitable size. After mixing thoroughly, a 1-pound composite sample shall be drawn for testing purposes. The composite sample shall be placed in an air-tight container, sealed, and marked with the identifying lot number and date.

4.4 Test procedures.

4.4.1 *Composition.* The composition (table I) shall be determined according to the methods specified in Fed. Test Method Std. No. 536 as indicated in table II. Compliance with the requirement that the soap powder contains no inert fillers shall be satisfied by a certificate of compliance from the supplier.

5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *One-pound quantities.* One pound of alkaline scrubbing soap shall be packaged in a sift-proof, folding paperboard box conforming to style I, type A, class a or b of PPP-B-566.

5.1.1.2 *One-hundred or 250-pound quantities.* One-hundred or 250 pounds of alkaline scrubbing soap shall be packaged in a fiber drum conforming to type III, grade A of PPP-D-723. Each drum shall be lined with a minimum 0.002-inch-thick polyethylene bag conforming to type I, grade A, finish 1 of L-P-378. All seams and closures shall be effected by heat sealing and shall conform to the heat seal strength requirement in L-P-378.

5.1.2 *Level B.*

TABLE II.—*Instructions for testing of the end item*

Characteristic	Specification reference		Results reported as	
	Requirement	Test method[1]	Pass or fail	Numerically to nearest[2]
Anhydrous soap	Table I	401		0.1 percent
Alkaline salts (calculated as Na_2CO_3)	Table I	304		0.1 percent
Sum of anhydrous soap and alkaline salts (alkaline salts calculated as Na_2CO_3)	Table I	—		0.1 percent
Matter insoluble in water	Table I	303		0.1 percent
Free alkali as NaOH	Table I	302		0.1 percent
Inert fillers	Table I	4.4.1		—

[1] All tests except for inert fillers are performed in accordance with Fed. Test. Method Std. No. 536.
[2] Test reports shall include all values on which results are based.

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5.1.2.1 *One-pound and 40-ounce quantities.* One pound or 40 ounces of alkaline scrubbing soap shall be packaged in sift-proof folding boxes as specified in 5.1.1.1.

5.1.2.2 *One-hundred or 250-pound quantities.* One-hundred or 250 pounds of alkaline scrubbing soap shall be packaged in a fiber drum conforming to type I, grade A of PPP-D-723. Each drum shall be lined with a polyethylene bag as specified in 5.1.1.2.

5.1.3 *Level C.* The alkaline scrubbing soap shall be packaged in accordance with the industry's practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Forty-eight 1-pound boxes of alkaline scrubbing soap packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style RSC of PPP-B-636. Each fiberboard shipping container shall be closed, waterproofed with tape, and reinforced with flat steel strapping or tape banding in accordance with appendix of the container specification. The drums shall not require overpacking.

5.2.2 *Level B.* As specified, forty-eight 1-pound boxes or twenty-four 40-ounce boxes of alkaline scrubbing soap packaged as specified in 5.1.2, shall be packed in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style RSC of PPP-B-636. All flaps of the fiberboard shipping container shall be securely sealed with an adhesive commercially used for the specific product being packed by application throughout the entire area of contact between the flaps, or by a combination of metal stitching the bottom flaps and sealing the top flaps with adhesive. The drums shall not require overpacking.

5.2.3 *Level C.* Alkaline scrubbing soap packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The alkaline scrubbing soap is a general purpose fatty acid soap containing alkaline salts. It is intended for heavy duty cleaning applications where high alkalinity is not objectionable. It is not intended for use on fabrics.

6.2 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Shade of color, when required (see 3.3).

(*c*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 *Level B.* Level B packaging is intended to provide economical but limited protection and should be specified only when it is known that the commodity will be ultimately used at a domestic activity.

6.3 **Color chips.** Color chips are available from all General Services Administration Business Service Centers.

6.4 **Basis of purchase.** The alkaline scrubbing soap shall be purchased by net weight

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as determined at the point of origin (manufacturer's plant) and certified as specified in the contract or order.

6.5 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Soap powders.
Carload minimum weight 36,000 pounds.

Motor:

Soap powder, not otherwise indexed.
Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Copies of this specification may be purchased for 5 cents each.

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